

Observationally Constrained Modeling of the Reactive Uptake of Isoprene-derived Epoxydiols under Elevated Relative Humidity and Varying Acidity of Seed Aerosol Conditions

Jie Zhang¹, Manish Shrivastava^{1,*}, Alla Zelenyuk¹, Rahul A. Zaveri¹, Jason D. Surratt^{2,3}, Matthieu Riva⁴, David Bell⁵, and Marianne Glasius⁶

1. Pacific Northwest National Laboratory, Richland, Washington 99352, United States
2. Department of Environmental Sciences and Engineering, Gillings School of Global Public Health, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina 27599, United States
3. Department of Chemistry, College of Arts and Sciences, University of North Carolina at Chapel Hill, Chapel Hill, North Carolina 27599, United States
4. Université Lyon, Université Claude Bernard Lyon 1, CNRS, IRCELYON, 69626, Villeurbanne, France
5. Laboratory of Atmospheric Chemistry, Paul Scherrer Institute, Villigen, 5232, Switzerland
6. Department of Chemistry, Aarhus University, Aarhus C, 8000, Denmark

* Correspondence to manishkumar.shrivastava@pnnl.gov

This supporting information contains 3 sections and 7 figures.

S1. Size-dependent loss of particles onto the chamber walls in the box model

In this study, the particle wall-loss module in the Aerosol Dynamics, gas- and particle-phase chemistry model for laboratory CHAMber studies (ADCHAM)¹ was implemented into our MOSAIC box model to simulate the particle loss onto the chamber walls. In this module, the loss of particles to chamber walls was analogous to particle migration in an effective electrical field and could be represented by a first-order deposition rate. The suspended particles in the chamber were assigned zero, one, two, and three elemental charges assuming a Boltzmann charge distribution in each size bin.

For zero-charge particles, the first-order deposition rate to the chamber walls was calculated following Lai and Nazaroff² by assuming that the turbulence in the chamber was homogeneous and isotropic. The Teflon chamber in Riva et al.'s experiments³ has dimensions of 1m×1m×1m. The deposition velocities onto the 4 vertical surfaces (k_{dv}), the upward horizontal surface (k_{du}), and the downward horizontal surface (k_{dd}) were calculated using Eq. (S1-S3):

$$k_{dv} = \frac{\mu^*}{I} \quad (\text{S1})$$

$$k_{du} = \frac{v_s}{1 - \exp\left(-\frac{v_s I}{\mu^*}\right)} \quad (\text{S2})$$

$$k_{dd} = \frac{v_s}{\exp\left(\frac{v_s I}{\mu^*}\right) - 1} \quad (\text{S3})$$

where μ^* is the friction velocity, which is adjustable to optimize the particle wall-loss in the model simulations best agreed with the chamber measurements. In this study, the $\mu^* = 0.12$ was derived from fitting particle loss for the experiment with acidified sulfate seed aerosols, and $\mu^* = 0.15$ was obtained for the experiment with non-acidified sulfate seed aerosols. I is a parameterized integral related to the flux of particles across the boundary layer and calculated using the equations in Table 2 from Lai and Nazaroff². v_s is the size-dependent terminal settling velocity of a particle with a diameter of D_p , which is calculated by Eq. (S4):

$$v_s = \frac{Dp^2 \rho g C_c}{18\mu} \quad (S4)$$

where μ is the dynamic viscosity of air in the chamber ($1.8 \times 10^{-5} \text{ kg m}^{-1} \text{ s}^{-1}$). C_c is the Cunningham slip correction factor, which is dependent on particle diameter (D_p) and gas mean free pathway (λ_i), calculated by Eq. (S5):

$$C_c = 1 + \frac{2\lambda_i}{D_p} \left(1.257 + 0.4 \exp \left(-\frac{1.1D_p}{2\lambda_i} \right) \right) \quad (S5)$$

Combining the deposition velocities calculated using Eq. (S1-S3), the overall deposition rate of non-charged particles, (s^{-1}) is calculated by Eq. (S6):

$$k_{non-charge} = \frac{k_{dv}A_v + k_{du}A_u + k_{dd}A_d}{V} \quad (S6)$$

where A_v , A_u , and A_d are the surface areas of the vertical surfaces (4 m^2), the upward-facing surface (1 m^2), and the downward-facing surface (1 m^2), respectively. V is the volume of the chamber (1 m^3).

For the charged particles, the first-order deposition rate, k_{charge} (s^{-1}) was calculated by Eq. (S7):

$$k_{charge} = \frac{AneC_cE}{V3\pi\mu D_p} \quad (S7)$$

where A is the total surface area of chamber walls (6 m^2); n is the number of elemental charges of the particle; e is the elementary charge; E is the mean electrical field strength. E is unknown and obtained of 10 V m^{-3} for both acidified and non-acidified experiments from optimization of simulated particle wall loss.

This size-dependent module has been evaluated by Xing et al.⁴ for a larger Teflon environmental chamber (10.6 m^3). In this study, with optimized parameters of μ^* and E , the simulated total suspended aerosol numbers show reasonable agreement with the continuous measurements during the chamber experiments, as shown in Figure S1.

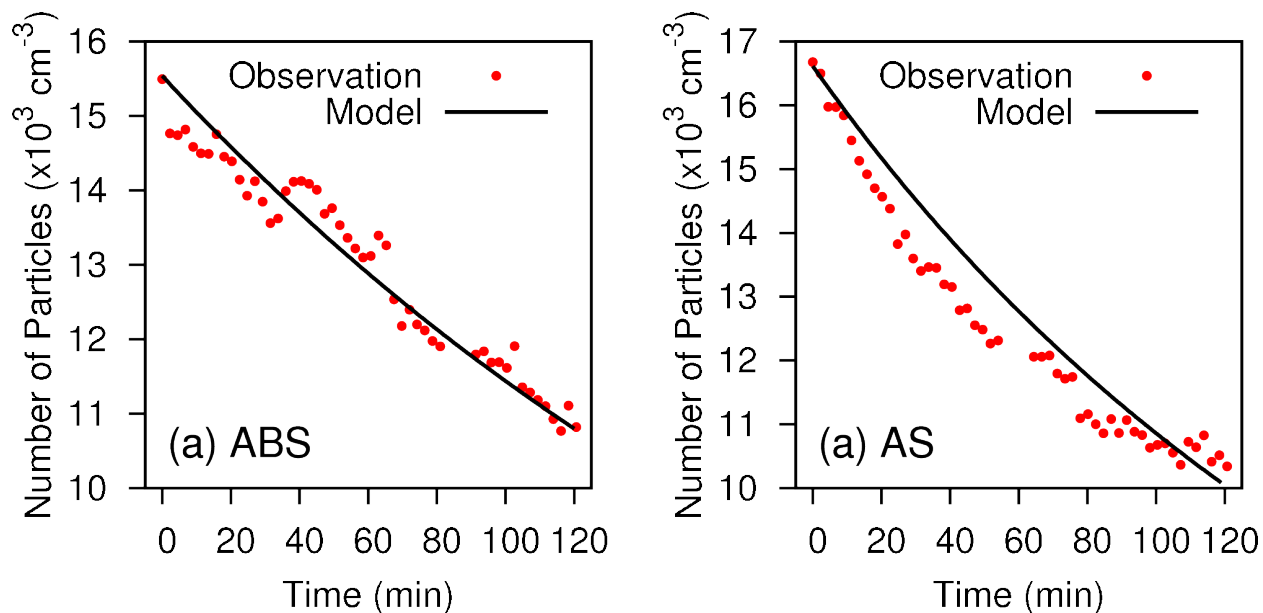


Figure S1. Time series of observed (red dots) and simulated (black line) total number concentrations for (a) acidified sulfate (ABS) aerosols and (b) non-acidified sulfate (AS) aerosols.

S2. Gas-phase chamber wall loss

The vapor wall loss in the Teflon chamber is treated as kinetically first-order uptake and release from the walls, as described in the previous publications⁵⁻⁷. The first-order uptake coefficient, $k_{w,on}$ (s^{-1}), is calculated based on Eq (S8):

$$k_{w,on} = \frac{2A}{\pi V} \sqrt{k_e D_v} \quad (S8)$$

where A/V is the ratio of total surface area to the volume for the chamber, k_e is the coefficient of eddy diffusion describing the mixing conditions in the chamber, which is calculated as $0.0096 s^{-1}$ for this 1-m³ Teflon chamber. D_v is the gas-phase diffusion coefficient of the vapor molecule, which is estimated as $0.079 cm^2 s^{-1}$ for IEPOX and $0.072 cm^2 s^{-1}$ for tetrols, depending on their molecular weight (MW) of vapors. The first-order release coefficient, $k_{w,off}$ (s^{-1}), is calculated following the Eq (S9):

$$k_{w,off} = k_{w,on} \frac{C^*}{C_w} \quad (S8)$$

76 where C^* is the saturation mass concentration ($10 \mu\text{g m}^{-3}$ for tetrols and $10^4 \mu\text{g m}^{-3}$ for IEPOX),
77 C_w is the equivalent organic mass concentration of the walls, which is a function of C^* and
78 calculated as $C_w = 16(C^*)^{0.6} \mu\text{g m}^{-3}$ ($C^* = 1 \sim 10^4 \mu\text{g m}^{-3}$)⁸. The $k_{w,on}$ and $k_{w,off}$ are estimated as
79 $1.003 \times 10^{-3} \text{ s}^{-1}$ and $1.575 \times 10^{-4} \text{ s}^{-1}$ for tetrols. The C^* of IEPOX is three orders of magnitude higher
80 than the tetrols, and its $k_{w,on}$ and $k_{w,off}$ are $1.052 \times 10^{-3} \text{ s}^{-1}$ and $2.617 \times 10^{-2} \text{ s}^{-1}$, respectively. The
81 impacts of gas-phase wall loss on the aerosol size evolution and SOA product compositions are
82 discussed in the main text *Effects of Tetrol Gas Wall Loss*.

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S3. Additional results

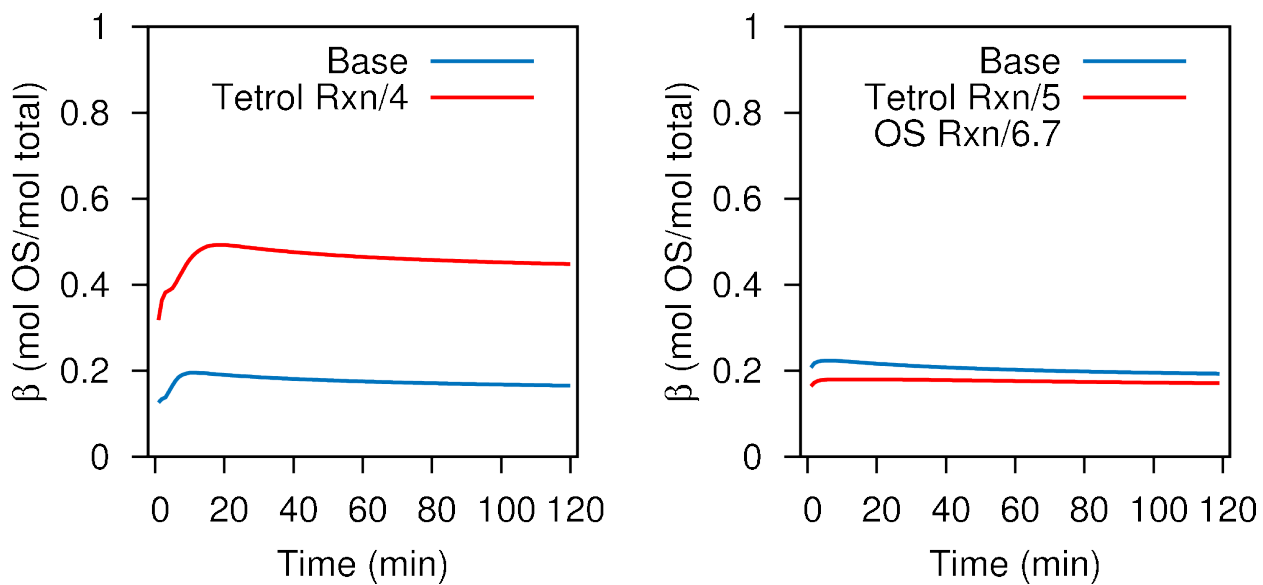


Figure S2. Time evolution of organosulfate (OS) branching ratio (β) for a selected aerosol bin with the initial midpoint diameter of 68.6 nm for the experiments with (a) ammonium bisulfate (ABS) and (b) ammonium sulfate (AS) aerosols.

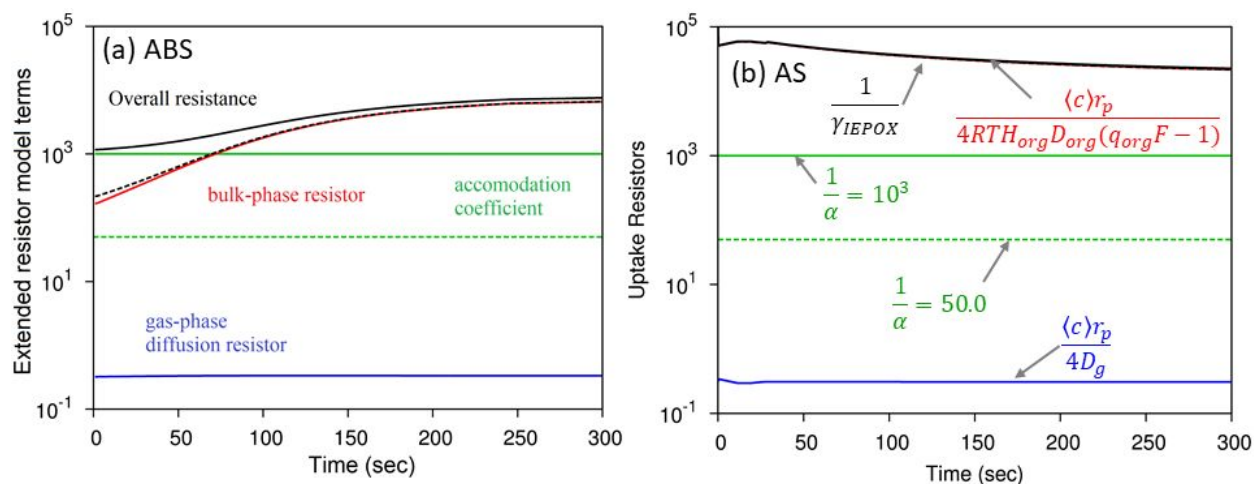


Figure S3. Model predicted overall resistance to mass transfer ($1/\gamma_{IEPOX}$) on (a) ABS and (b) AS seeds consists of three resistor model terms: gas-phase diffusion resistor (blue), accommodation coefficient (green), and bulk-phase resistor (red). The $1/\gamma_{IEPOX}$ (black) is the sum of three resistor terms. These results are based on the coefficients in the aerosol bin with initial diameter of 68.6 nm.

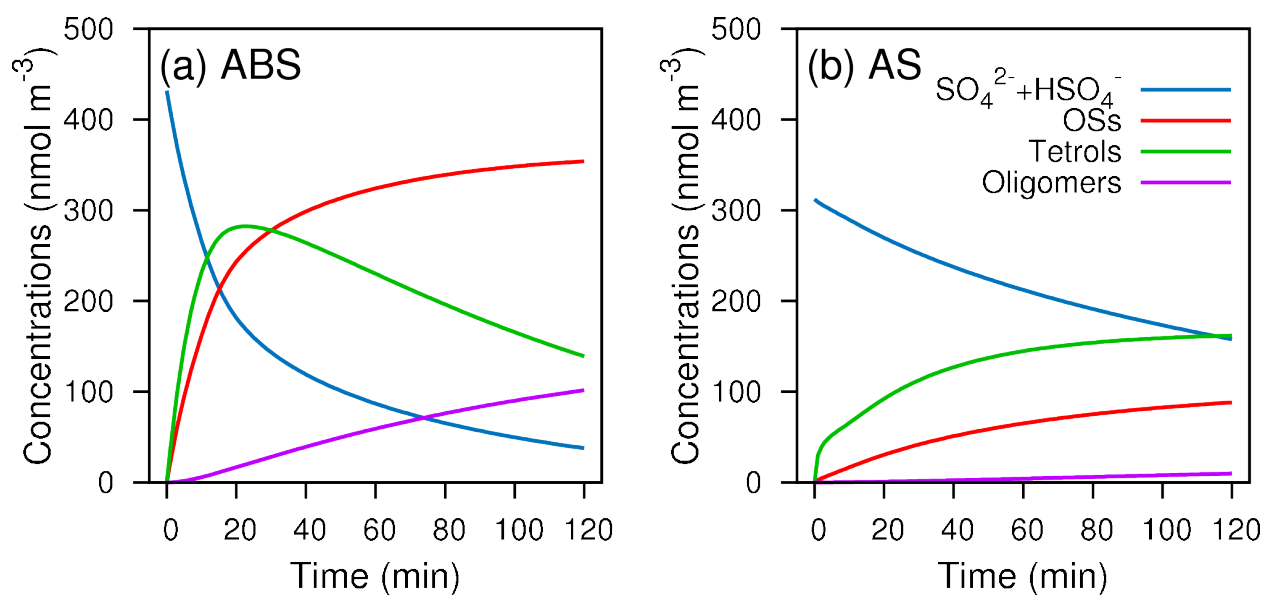


Figure S4. Time series concentrations (nmol m⁻³) of inorganic sulfates ($\text{SO}_4^{2-} + \text{HSO}_4^-$), OSs, tetrols, and oligomers in the chamber for (a) ABS seed experiment and (b) AS seed experiment.

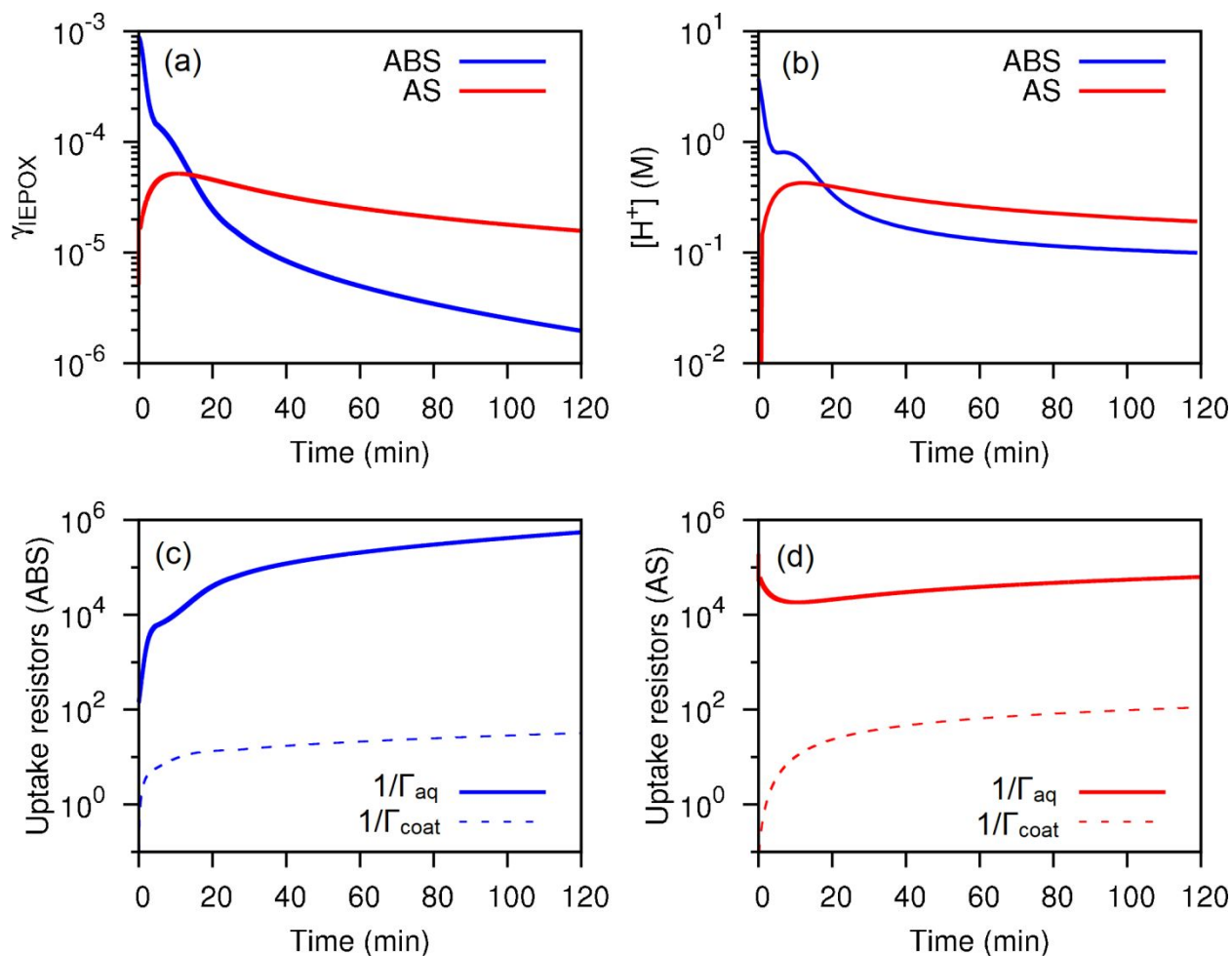


Figure S5. Time evolution of (a) uptake coefficient of IEPOX (γ_{IEPOX}), (b) H^+ concentration, and reaction resistor (Γ_{aq}) and coating effect resistor (Γ_{coat}) for (c) ABS and (d) AS simulations in the aerosol bin with initial diameter of 68.6 nm.

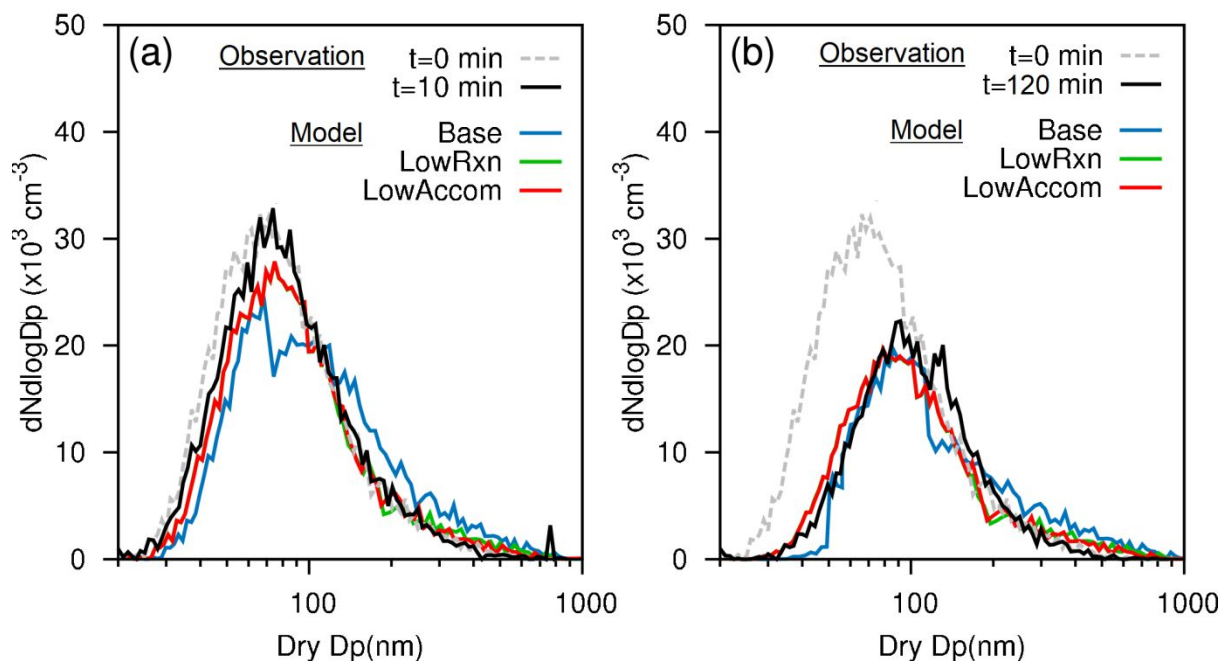


Figure S6. Size distribution evolution of aerosol number concentration of observations (black lines) and model simulations (color lines) at (a) 10 min and (b) 120 min from the start of introducing IEPOX gas (grey dashed lines) onto AS aerosols. The overprediction of particle distribution in the large size bins leads to the higher model simulated total volume concentration in Figure 2d.

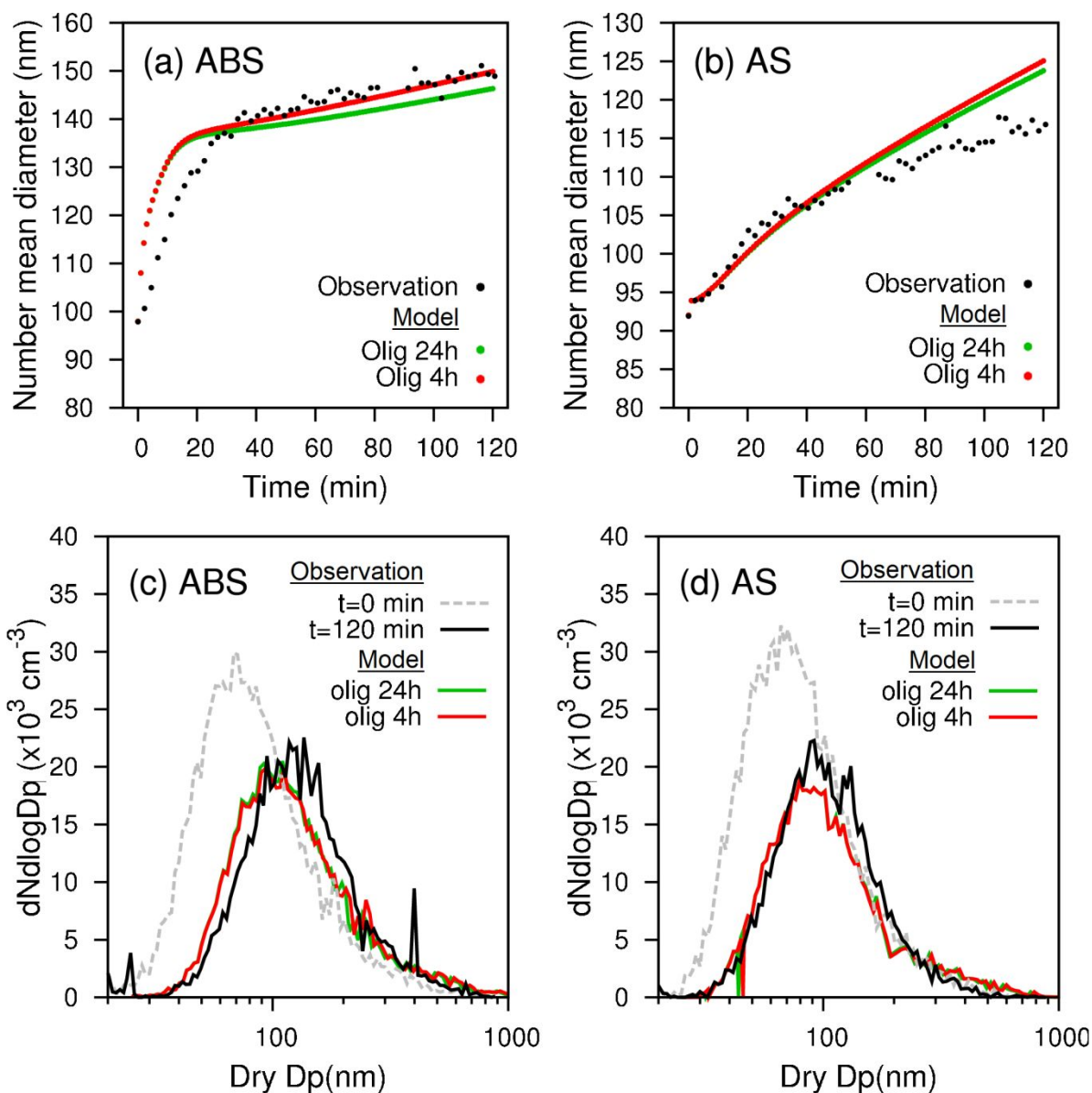


Figure S7. Time evolution of number mean diameter during the IEPOX uptake on (a) ABS seeds and (b) AS seeds and aerosol size distribution at 120 min for (c) ABS seeds and (d) AS seeds with different oligomerization time scale of 4 h and 24 h.

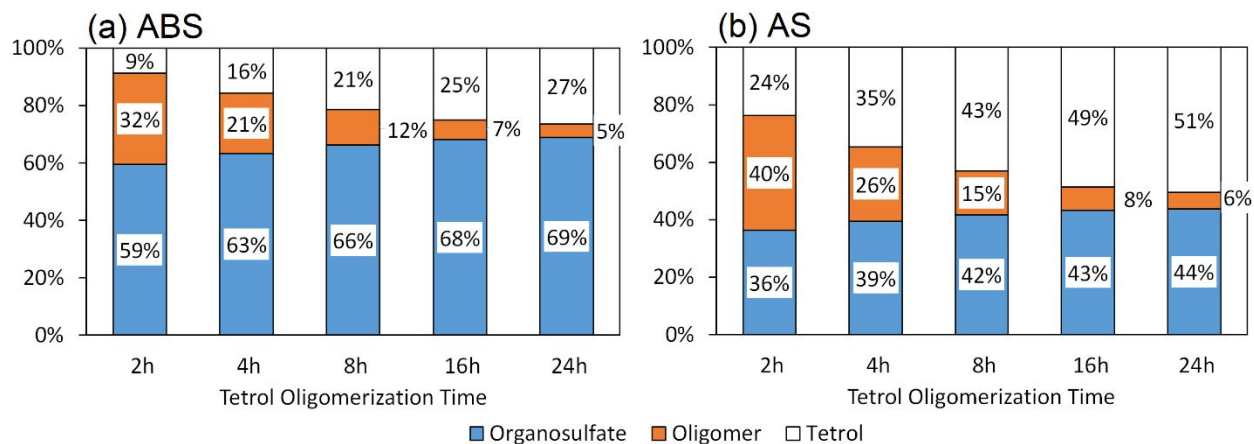


Figure S8. Sensitivity of relative mass fractions of simulated IEPOX-SOA compositions (OS, tetrols, and oligomers) at the end of experiments on tetrol oligomerization timescale (2 – 24 h) with (a) ABS seeds and (b) AS seeds.

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